GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 1 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 2	Lesson: 1
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Integers - Identification of integers Integers - Representatio n of integers on number line

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Define integers and distinguish them from non-integers - Identify positive integers, negative integers and zero in different situations - Appreciate the use of integers in daily life situations

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we identify integers in real life situations?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 1 - Thermometer s - Number cards - Charts with integers - Master Mathematics Grade 8, pg. 2 - Manila paper - Rulers - Markers - Number lines

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Discuss and find readings of thermometers showing positive and negative values - Classify numbers as integers or non-integers - Use real-life situations like floors above and below ground to represent integers

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Oral questions - Written exercises

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 2 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 2	Lesson: 2
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Integers - Addition of integers on number line Integers - Subtraction of integers on number line

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - State the rule for adding integers on a number line - Carry out addition of integers on a number line correctly - Value the importance of addition of integers in real life

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we carry out addition of integers?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 3 - Number cards - Ground markings - Chalk - Counters - Master Mathematics Grade 8, pg. 4 - Number lines - Markers - Playground space

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Use number cards and containers for selection - Draw number lines on the ground - Jump to the right to add positive numbers - Mark and record positions after jumping

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Practical activities - Oral questions

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 3 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 2	Lesson: 3
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Integers - Combined operations on number line Integers - Application of integers using IT resources

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Describe the order of combined operations on integers - Perform combined addition and subtraction of integers on number line - Show confidence in solving problems involving integers

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we perform combined operations of integers?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 5 - Number lines - Temperature gauges - Real- life problem cards - Master Mathematics Grade 8, pg. 6 - Digital devices - Internet access - Integer games/apps

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Practice mixed operations using number lines - Solve problems involving temperature changes - Work out problems involving floors in buildings

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written exercises - Problem- solving tasks - Observation

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 4 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 2	Lesson: 4
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Fractions - Order of operations in fractions

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Identify the correct order of operations in fractions (BODMAS) - Carry out combined operations on fractions accurately - Appreciate the importance of following correct order

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we use fractions in real life situations?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 8 - Fraction cards - Calculators - Charts showing BODMAS

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Discuss and use correct order of operations in fractions - Work through operations in brackets first, then multiplication/ division, then addition/subtr action - Copy operations on number cards and solve

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written tests - Class activities - Oral questions

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 5 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 2	Lesson: 5
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Fractions - Operations on fractions from shopping activities Fractions - Word problems involving fractions

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain how fractions are used in shopping and trading - Work out fraction operations from shopping activities - Show responsibility in applying fractions to real situations

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

Where do we apply combined operations on fractions?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 9 - Shopping lists - Price tags - Play money - Fraction pieces - Master Mathematics Grade 8, pg. 10 - Word problem cards - Fraction charts - Measuring tools

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Discuss and carry out operations on fractions from shopping and other real-life cases - Role- play shopping scenarios - Solve problems involving sharing and distribution

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Problem- solving - Practical activities - Written assignments

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 6 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 3	Lesson: 1
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Fractions - Games and IT activities on fractions

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Describe different games involving fractions - Use IT devices for learning operations on fractions and play games - Enjoy learning about fractions

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How can we make learning fractions more interesting?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 11 - Tablets/comp uters - Internet access - Fraction games

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Play games of operations on fractions using IT devices or other resources - Engage in drag and drop activities online - Use fraction apps for practice

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Game performance - Digital assessment

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 7 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 3	Lesson: 2
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Fractions - Mixed practice on combined operations

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Recall the order of operations in fractions - Solve complex combined fraction operations proficiently - Show confidence in working with fractions

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

What strategies help us solve complex fraction problems?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 12 - Exercise books - Fraction worksheets - Group work materials

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Practice solving mixed fraction problems - Work in groups on challenging fraction tasks - Present solutions to the class

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written tests - Group presentations - Peer assessment

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 8 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 3	Lesson: 3
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Fractions - Mixed practice on combined operations

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Recall the order of operations in fractions - Solve complex combined fraction operations proficiently - Show confidence in working with fractions

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

What strategies help us solve complex fraction problems?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 12 - Exercise books - Fraction worksheets - Group work materials

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Practice solving mixed fraction problems - Work in groups on challenging fraction tasks - Present solutions to the class

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written tests - Group presentations - Peer assessment

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 9 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 3	Lesson: 4
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Fractions - Application and reflection

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Discuss various applications of fractions in daily life - Demonstrate mastery of fraction operations - Promote use of fractions in real life situations

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How have fractions helped us in our daily lives?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 13 - Portfolio materials - Reflection journals

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Discuss with peers, parents or guardians about areas where fractions are applied - Share real-life experiences involving fractions - Compile a portfolio of fraction work

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Portfolio assessment - Oral presentations - Self- assessment

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 10 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 3	Lesson: 5
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Decimals - Conversion of fractions to decimals

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain the relationship between fractions and decimals - Convert fractions to decimals using different methods - Appreciate the connection between fractions and decimals

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we convert fractions to decimals?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 13 - Conversion charts - Calculators - Place value charts

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Practice converting fractions to decimals using equivalent fractions with denominators as powers of 10 - Use long division method to convert fractions - Complete conversion tables

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written exercises - Oral questions - Class activities

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 11 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 4	Lesson: 1
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Decimals - Identifying and converting recurring decimals

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Define recurring and non-recurring decimals - Identify recurring decimals and convert them to fractions correctly - Show interest in working with recurring decimals

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we identify and work with recurring decimals?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 15 - Decimal cards - Number cards - Calculators

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Discuss and classify non- recurring and recurring decimals - Indicate recurring digits using dot notation - Practice converting recurring decimals to fractions using algebraic method

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written tests - Practical exercises - Observation

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 12 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 4	Lesson: 2
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Decimals - Rounding off decimals to decimal places

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - State the rules for rounding off decimals - Round off decimal numbers to required decimal places accurately - Value accuracy in rounding decimals

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we round off decimals correctly?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 19 - Place value charts - Decimal number cards - Rounding worksheets

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Discuss and round off decimal numbers to required decimal places - Practice rounding to 1, 2, 3 decimal places - Use place value charts to understand rounding

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written assignments - Oral questions - Class tests

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 13 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 4	Lesson: 3
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Decimals - Expressing numbers in significant figures

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain the meaning of significant figures - Write decimal and whole numbers to given significant figures - Show precision in expressing numbers

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

What are significant figures and why are they important?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 21 - Number charts - Worksheets - Scientific calculators

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Write decimal and whole numbers to given significant figures - Discuss rules for identifying significant figures - Practice expressing numbers to different significant figures

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written tests - Practical exercises - Observation

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 14 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 4	Lesson: 4
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Decimals - Expressing numbers in standard form

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Define standard form notation A × 10■ - Write numbers in standard form correctly and convert them back - Appreciate the use of standard form for very large and small numbers

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we express numbers in standard form?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 23 - Standard form cards - Calculators - Charts

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Write numbers in standard form on learning materials such as cards or charts - Practice expressing very large and very small numbers - Understand the power of 10 notation

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written exercises - Oral questions - Class activities

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 15 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 4	Lesson: 5
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Decimals - Combined operations on decimals

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Identify the correct order of operations for decimals - Work out combined operations on decimals systematically - Show confidence in solving decimal problems

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we perform combined operations on decimals?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 24 - Operation cards - Calculators - Worksheets

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Work out combined operations on decimals in the correct order - Practice problems involving brackets, multiplication, division, addition and subtraction - Solve complex decimal calculations

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written tests - Problem- solving - Observation

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 16 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 5	Lesson: 1
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Decimals - Application of decimals to real life

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Identify situations where decimals are used in daily life - Apply decimals to solve practical problems - Promote use of decimals in daily activities

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

Where do we use decimals in our daily lives?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 26 - Real-life problem cards - Measuring instruments - Price lists

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Discuss and apply decimals to real life cases - Solve problems involving money, measurement s, temperature - Work with real-life scenarios

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Practical tasks - Written assignments - Oral presentations

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 17 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 5	Lesson: 2
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Decimals - Application of decimals to real life

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Identify situations where decimals are used in daily life - Apply decimals to solve practical problems - Promote use of decimals in daily activities

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

Where do we use decimals in our daily lives?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 26 - Real-life problem cards - Measuring instruments - Price lists

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Discuss and apply decimals to real life cases - Solve problems involving money, measurement s, temperature - Work with real-life scenarios

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Practical tasks - Written assignments - Oral presentations

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 18 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 5	Lesson: 3
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Decimals - Games and digital activities

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain how digital games enhance learning of decimals - Use IT devices to play games involving decimals - Enjoy learning about decimals through interactive activities

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How can technology enhance our understanding of decimals?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 27 - Digital devices - Decimal games/apps - Internet access

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Play games of operations on decimals using IT or other materials - Use decimal apps and online games - Engage in interactive decimal activities

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Game performance - Participation

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 19 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 5	Lesson: 4
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Squares and Square Roots - Reading squares from tables

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain how to read mathematical tables for squares - Work out squares of numbers between 1.0 and 9.999 from tables - Show accuracy in using mathematical tables

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

What are squares of numbers?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 29 - Mathematical tables - Number cards - Worksheets

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Read and write the squares of numbers from tables - Practice locating numbers in the table and reading their squares - Work through examples using Table 1.3

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Practical exercises - Written tests - Observation

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 20 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 5	Lesson: 5
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Squares and Square Roots - Squares of large numbers

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Describe the method for finding squares of numbers above 10 - Work out squares of numbers above 10 using standard form and tables - Demonstrate systematic approach in calculations

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we find squares of numbers greater than 10?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 33 - Mathematical tables - Standard form charts - Calculators

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Practice finding squares of numbers above 10 using standard form method - Convert numbers to standard form A × 10■ - Calculate squares and express in ordinary form

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written exercises - Class activities - Oral questions

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 21 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 6	Lesson: 1
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Squares and Square Roots - Squares of numbers less than 1

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain the process for squaring decimal numbers less than 1 - Find squares of decimal numbers less than 1 using tables - Show precision in working with small numbers

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we find squares of numbers less than 1?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 35 - Mathematical tables - Decimal cards - Worksheets

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Practice finding squares of numbers less than 1 - Use standard form with negative powers of 10 - Apply systematic method for calculations

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written tests - Practical exercises - Problem- solving

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 22 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 6	Lesson: 2
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Squares and Square Roots - Reading square roots from tables

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain how to read square root tables - Work out square roots of numbers from 1 to 99.99 using tables - Appreciate the relationship between squares and square roots

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

Where do we apply square roots in real life?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 37 - Mathematical tables - Square root charts - Number cards

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Read and write the square roots of numbers from tables - Practice using Table 1.4 for square roots - Add values from the ADD column correctly

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written assignments - Oral questions - Class tests

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 23 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 6	Lesson: 3
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Squares and Square Roots - Reading square roots from tables

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain how to read square root tables - Work out square roots of numbers from 1 to 99.99 using tables - Appreciate the relationship between squares and square roots

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

Where do we apply square roots in real life?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 37 - Mathematical tables - Square root charts - Number cards

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Read and write the square roots of numbers from tables - Practice using Table 1.4 for square roots - Add values from the ADD column correctly

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written assignments - Oral questions - Class tests

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 24 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 6	Lesson: 4
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Squares and Square Roots - Square roots of large numbers

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Describe the method for finding square roots of numbers 100 and above - Find square roots of numbers 100 and above using tables - Show systematic approach in calculations

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we find square roots of numbers above 100?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 39 - Mathematical tables (Tables 1.4 & 1.5) - Worksheets - Calculators

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Practice finding square roots of numbers above 100 - Use standard form method - Work with both Table 1.4 and Table 1.5 appropriately

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written exercises - Practical work - Observation

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 25 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 6	Lesson: 5
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Squares and Square Roots - Using calculators for squares and square roots

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Identify the square and square root functions on a calculator - Work out squares and square roots using a calculator correctly - Appreciate the efficiency of using calculators

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do calculators help us find squares and square roots?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 42 - Scientific calculators - Digital devices - Comparison worksheets

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Practice working out squares and square roots using a calculator - Compare calculator results with table results - Use IT devices or other materials to play square and square root games

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Practical exercises - Observation - Written tests

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 26 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 7	Lesson: 1
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Rates, Ratio, Proportions and Percentages - Identifying rates

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Define rate as a quotient relationship between two quantities - Identify rates in different real-life situations - Appreciate the use of rates in daily life

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we use rates in real life situations?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 44 - Stopwatches - Rate cards - Mobile phones (for demonstration)

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Time while doing different activities such as calling using different mobile service providers - Role play activities and note time taken - Record and compare rates

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Oral questions - Practical activities

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 27 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 7	Lesson: 2
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Rates, Ratio, Proportions and Percentages - Working out rates

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain the method for calculating rates - Calculate rates from given information accurately - Show precision in rate calculations

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we calculate rates from given information?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 46 - Timers - Measuring tools - Rate worksheets

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Carry out activities to determine rates - Calculate rates per unit time or quantity - Solve rate problems from real-life contexts

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written tests - Problem- solving - Class activities

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 28 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 7	Lesson: 3
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Rates, Ratio, Proportions and Percentages - Expressing fractions as ratios

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain how to convert fractions to ratios - Express fractions as ratios in simplest form - Value precision in ratio work

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we express fractions as ratios?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 48 - Cut-out materials - Ratio cards - Counters

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Use cut outs from whole objects to relate fractions to ratios - Practice writing fractions as numerator : denominator - Simplify ratios to lowest terms

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written exercises - Practical work - Oral questions

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 29 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 7	Lesson: 4
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Rates, Ratio, Proportions and Percentages - Comparing ratios

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Describe methods for comparing two or more ratios - Compare ratios using percentage method and LCM method - Show systematic approach in comparing ratios

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we compare two or more ratios?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 50 - Comparison charts - Ratio cards - Calculators

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Discuss and compare ratios from cut outs - Use LCM method to compare ratios - Express ratios as percentages for easy comparison

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written tests - Class activities - Problem- solving

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 30 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 7	Lesson: 5
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Rates, Ratio, Proportions and Percentages - Comparing ratios

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Describe methods for comparing two or more ratios - Compare ratios using percentage method and LCM method - Show systematic approach in comparing ratios

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we compare two or more ratios?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 50 - Comparison charts - Ratio cards - Calculators

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Discuss and compare ratios from cut outs - Use LCM method to compare ratios - Express ratios as percentages for easy comparison

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written tests - Class activities - Problem- solving

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 31 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 8	Lesson: 1
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Rates, Ratio, Proportions and Percentages - Division of quantities in ratios

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain the process of dividing quantities in given ratios - Divide quantities in given ratios systematically - Show fairness in sharing quantities

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we divide quantities using ratios?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 51 - Counters - Bottle tops - Sharing materials

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Discuss and share quantities of concrete objects in different ratios - Use counters or bottle tops to practice sharing - Solve sharing problems

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Practical exercises - Written assignments - Observation

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 32 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 8	Lesson: 2
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Rates, Ratio, Proportions and Percentages - Working out ratios

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Identify the method for finding ratios from given quantities - Work out ratios in different situations - Appreciate applications of ratios in daily life

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we determine ratios from given information?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 53 - Data cards - Real- life examples - Worksheets

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Calculate ratios from given quantities - Find missing values in ratio problems - Apply ratios to real situations

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written tests - Problem- solving - Oral questions

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 33 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 8	Lesson: 3
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Rates, Ratio, Proportions and Percentages - Increase and decrease using ratios

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain how ratios show increase or decrease in quantities - Work out increase and decrease of quantities using ratios - Apply ratio changes to real situations

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do ratios represent increase or decrease?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 55 - Change scenario cards - Calculators - Worksheets

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Discuss and determine increase and decrease using ratios - Use the format new : old to express changes - Solve problems involving ratio changes

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written exercises - Class activities - Problem- solving

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 34 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 8	Lesson: 4
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Rates, Ratio, Proportions and Percentages - Percentage increase

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Define percentage increase - Calculate percentage increase accurately using the formula - Show precision in percentage calculations

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we calculate percentage increase?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 57 - Percentage charts - Calculators - Problem cards

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Discuss and determine percentage increase of different quantities - Use the formula: percentage change = (change/origin al) × 100% - Solve real-life percentage problems

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written tests - Practical exercises - Oral questions

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 35 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 8	Lesson: 5
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Rates, Ratio, Proportions and Percentages - Percentage decrease

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Define percentage decrease - Calculate percentage decrease correctly - Apply percentage decrease to real situations responsibly

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we calculate percentage decrease?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 58 - Discount cards - Price lists - Calculators

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Work through percentage decrease problems - Calculate new values after percentage decrease - Solve problems involving discounts and reductions

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written assignments - Problem- solving - Class tests

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 36 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 9	Lesson: 1
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Rates, Ratio, Proportions and Percentages - Identifying direct proportions

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Define direct proportion - Identify direct proportions in real life situations - Appreciate proportional relationships in daily activities

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

What is direct proportion?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 59 - Proportion charts - Real- life examples - Digital devices

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Use IT devices or other materials to explore proportions - Role play shopping activities to show direct relationships - Identify situations where increase in one leads to increase in other

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Oral questions - Practical activities

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 37 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 9	Lesson: 2
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Rates, Ratio, Proportions and Percentages - Working out direct proportions

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain the unitary method for solving direct proportion - Work out direct proportions systematically - Show accuracy in direct proportion calculations

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we solve direct proportion problems?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 60 - Proportion tables - Worksheets - Calculators

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Complete tables showing direct proportional relationships - Calculate missing values in direct proportion - Apply direct proportion to solve problems

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written tests - Problem- solving - Class activities

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 38 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 9	Lesson: 3
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Rates, Ratio, Proportions and Percentages - Working out direct proportions

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain the unitary method for solving direct proportion - Work out direct proportions systematically - Show accuracy in direct proportion calculations

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we solve direct proportion problems?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 60 - Proportion tables - Worksheets - Calculators

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Complete tables showing direct proportional relationships - Calculate missing values in direct proportion - Apply direct proportion to solve problems

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written tests - Problem- solving - Class activities

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 39 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 9	Lesson: 4
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Rates, Ratio, Proportions and Percentages - Identifying indirect proportions

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Define indirect proportion - Identify indirect proportions in different situations - Appreciate the difference between direct and indirect proportion

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

What is indirect proportion?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 62 - Hourglass - Containers - Bottle tops

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Use hourglass to show and determine indirect relationships - Identify situations where increase in one leads to decrease in other - Practice with filling containers

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Practical work - Oral questions

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 40 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 9	Lesson: 5
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Rates, Ratio, Proportions and Percentages - Working out indirect proportions

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain the method for solving indirect proportion - Work out indirect proportions systematically - Show understanding of inverse relationships

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we solve indirect proportion problems?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 63 - Proportion worksheets - Calculators - Problem cards

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Complete tables showing indirect proportional relationships - Calculate values where ratios are inverted - Solve time- speed- distance problems

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written exercises - Problem- solving - Written tests

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 41 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 10	Lesson: 1
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Rates, Ratio, Proportions and Percentages - Application and reflection

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Discuss various applications of ratios and proportions - Apply ratios and proportions in various real- life contexts - Promote use of ratios and proportions in real life

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do ratios and proportions help us in daily life?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 64 - Video resources - Digital devices - Portfolio materials

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Watch videos on ratios and proportions as used in daily activities - Discuss applications with parents or guardians - Reflect on learning and compile portfolio

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Portfolio assessment - Presentations - Self- assessment

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 42 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 10	Lesson: 2
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Algebra
Sub-strand / lesson focus: Algebraic Expressions - Factorisation of algebraic expressions

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Define factorisation as the reverse of expansion - Identify the highest common factor (HCF) in algebraic expressions - Appreciate the use of factorisation in simplifying expressions

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we factorise algebraic expressions?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 65 - Number cards - Algebraic expression cards - Charts

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Make three sets of cards showing algebraic expressions and their factored forms - Match cards from different rows to form equations - Discuss and identify common factors in terms - Write HCF in front of brackets and remaining factors inside

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Card matching activity - Oral questions

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 43 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 10	Lesson: 3
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Algebra
Sub-strand / lesson focus: Algebraic Expressions - Identifying like and unlike terms in factorisation

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain the concept of like and unlike terms - Find common factors for different sets of terms - Show systematic approach in identifying factors

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

What makes terms like or unlike in algebra?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 67 - Factor cards - Worksheets - Group work materials

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Discuss and identify like and unlike terms - Find common factors from given sets of algebraic terms - Practice factorising expressions with numerical and variable common factors - Work in groups to factorise various expressions

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written exercises - Group presentations - Class activities

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 44 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 10	Lesson: 4
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Algebra
Sub-strand / lesson focus: Algebraic Expressions - Simplification of algebraic fractions

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain the process of simplifying algebraic fractions - Simplify algebraic fractions by finding LCM of denominators - Value accuracy in simplifying fractions

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we simplify algebraic expressions?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 68 - Fraction charts - LCM charts - Worksheets

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Discuss like and unlike terms in algebraic fractions - Find LCM of denominators in algebraic fractions - Combine fractions with different denominators - Practice simplifying complex algebraic fractions

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written tests - Practical exercises - Problem- solving

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 45 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 10	Lesson: 5
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Algebra
Sub-strand / lesson focus: Algebraic Expressions - Simplification of algebraic fractions

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain the process of simplifying algebraic fractions - Simplify algebraic fractions by finding LCM of denominators - Value accuracy in simplifying fractions

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we simplify algebraic expressions?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 68 - Fraction charts - LCM charts - Worksheets

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Discuss like and unlike terms in algebraic fractions - Find LCM of denominators in algebraic fractions - Combine fractions with different denominators - Practice simplifying complex algebraic fractions

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written tests - Practical exercises - Problem- solving

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 46 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 11	Lesson: 1
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Algebra
Sub-strand / lesson focus: Algebraic Expressions - Advanced simplification practice

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Describe steps for simplifying complex algebraic fractions - Simplify algebraic fractions involving multiple operations - Show confidence in working with algebraic fractions

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

What strategies help us simplify complex algebraic fractions?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 69 - Practice worksheets - Real-life problem cards - Calculators

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Practice writing fractions as single fractions - Simplify fractions with algebraic denominators - Solve problems involving algebraic fractions - Work through real-life applications

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written assignments - Class tests - Oral questions

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 47 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 11	Lesson: 2
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Algebra
Sub-strand / lesson focus: Algebraic Expressions - Using IT devices and application

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Identify IT resources for learning algebra - Use IT devices to work out algebra exercises and drag-drop activities - Enjoy using algebraic expressions in real life situations

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How can technology enhance our understanding of algebra?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 71 - Digital devices - Internet access - Algebra apps/software

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Use IT devices to work out exercises and activities in algebra - Engage in drag and drop activities of grouping similar terms - Play online games simplifying algebraic expressions - Discuss applications with peers and parents

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Digital assessment - Participation

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 48 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 11	Lesson: 3
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Algebra
Sub-strand / lesson focus: Linear Equations - Forming linear equations in two unknowns

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Define linear equations in two unknowns - Form linear equations from real-life situations using two variables - Show interest in forming equations from word problems

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we solve linear equations in two unknowns?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 72 - Beam balance - Masses (500g) - Marbles - Shopping scenario cards

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Put masses on beam balance and add marbles to balance - Give letters to represent unknowns - Role play shopping activities to form equations - Write equations from balancing scenarios

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Practical activities - Oral questions

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 49 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 11	Lesson: 4
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Algebra
Sub-strand / lesson focus: Linear Equations - More practice on forming equations

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Interpret word problems involving two unknowns - Form linear equations from various real-life scenarios - Appreciate the relevance of equations in daily life

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

Where do we use linear equations in two unknowns in real life situations?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 73 - Word problem cards - Real-life scenario cards - Worksheets

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Write equations to represent ages, costs, and quantities - Form equations from perimeter problems - Create equations from problems involving animals and farming - Practice with two-digit number problems

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written exercises - Problem- solving - Class activities

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 50 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 11	Lesson: 5
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Algebra
Sub-strand / lesson focus: Linear Equations - Solving by substitution method

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain the substitution method for solving simultaneous equations - Solve linear equations in two unknowns using substitution systematically - Show precision in solving equations

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we use substitution method to solve linear equations?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 74 - Fruit pictures - Equation cards - Step-by-step charts

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Write equations from fruit vendor scenario - Name equations as (i) and (ii) - Write one variable in terms of another - Replace and simplify to find values of unknowns

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written tests - Practical exercises - Oral questions

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 51 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 12	Lesson: 1
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Algebra
Sub-strand / lesson focus: Linear Equations - Advanced practice on substitution method

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Describe the complete process of substitution method - Solve complex simultaneous equations by substitution - Demonstrate mastery of substitution technique

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

What are the key steps in substitution method?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 75 - Practice worksheets - Real-life problem cards - Calculators

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Practice solving equations with fractions using substitution - Work through problems involving costs and quantities - Solve problems about carpentry and furniture making - Apply substitution to number problems

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written assignments - Problem- solving - Class tests

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 52 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 12	Lesson: 2
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Algebra
Sub-strand / lesson focus: Linear Equations - Advanced practice on substitution method

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Describe the complete process of substitution method - Solve complex simultaneous equations by substitution - Demonstrate mastery of substitution technique

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

What are the key steps in substitution method?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 75 - Practice worksheets - Real-life problem cards - Calculators

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Practice solving equations with fractions using substitution - Work through problems involving costs and quantities - Solve problems about carpentry and furniture making - Apply substitution to number problems

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written assignments - Problem- solving - Class tests

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 53 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 12	Lesson: 3
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Algebra
Sub-strand / lesson focus: Linear Equations - Solving by elimination method

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain the elimination method for solving simultaneous equations - Solve linear equations using elimination method systematically - Appreciate the efficiency of elimination method

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we solve equations using elimination method?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 76 - Shopping scenario cards - Elimination charts - Step- by-step guides

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Form equations from shopping scenarios (plates and cups) - Multiply equations to make coefficients equal - Subtract corresponding parts to eliminate one variable - Solve for remaining variable and substitute back

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written exercises - Practical work - Oral questions

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 54 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 12	Lesson: 4
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Algebra
Sub-strand / lesson focus: Linear Equations - More practice on elimination method

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Identify when to use elimination method - Solve various simultaneous equations by elimination efficiently - Show confidence in choosing appropriate methods

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

When is elimination method more suitable than substitution?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 78 - Comparison charts - Practice worksheets - Method selection guides

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Practice solving equations involving bread and tea leaves - Work through problems with different coefficients - Solve problems about costs of items - Compare elimination and substitution methods

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written tests - Class activities - Problem- solving

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 55 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 12	Lesson: 5
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Algebra
Sub-strand / lesson focus: Linear Equations - Application in real-life situations

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Discuss various applications of linear equations in daily life - Apply linear equations to solve real-life problems involving rectangles, costs, and quantities - Recognize use of linear equations in real life

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do linear equations help us solve real-life problems?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 79 - Video resources - Real-life scenario cards - Digital devices - Application worksheets

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Find sum and difference of two numbers using equations - Solve problems about rectangular flower beds - Work out problems involving hiring labourers - Apply equations to school fees and shopping scenarios - Watch videos on linear equations applications

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Portfolio assessment - Presentations - Written assignments - Self- assessment

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 1 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 2	Lesson: 1
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.3: Scale Drawing - Representatio n of length to given scale

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Define scale and its purpose - Determine scale from given measurement s - Show understanding of proportion

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

Why do we need scale when drawing large objects?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Ruler - Tape measure - Pencil - Drawing paper

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Compare sizes of objects and their representation s - Discuss need for scale in drawings - Measure actual dimensions - Choose appropriate scale for representation s - Calculate scale from given information - Express scale in different forms

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Oral questions - Practical tasks

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 2 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 2	Lesson: 2
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.3: Scale Drawing - Converting actual length to scale length

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - State the formula for converting actual length to scale length - Convert actual measurement s to scale measurement s accurately - Demonstrate computational skills

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we calculate scale length from actual length?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Calculator - Ruler - Conversion tables - Pencil

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Apply given scales to convert measurement s - Complete tables converting actual to scale lengths - Calculate scale lengths using various scales - Work with different units - Practice systematic conversions

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Written assignments - Problem- solving

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 3 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 2	Lesson: 3
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.3: Scale Drawing - Converting actual length to scale length

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - State the formula for converting actual length to scale length - Convert actual measurement s to scale measurement s accurately - Demonstrate computational skills

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we calculate scale length from actual length?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Calculator - Ruler - Conversion tables - Pencil

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Apply given scales to convert measurement s - Complete tables converting actual to scale lengths - Calculate scale lengths using various scales - Work with different units - Practice systematic conversions

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Written assignments - Problem- solving

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 4 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 2	Lesson: 4
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.3: Scale Drawing - Converting scale length to actual length

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain the process of converting scale to actual measurement s - Convert scale measurement s to actual measurement s accurately - Show systematic calculation approach

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we find real dimensions from scale drawings?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Ruler - Calculator - Scale drawings - Pencil

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Measure lengths on scale diagrams - Use given scales to find actual lengths - Calculate actual distances - Work with different unit conversions - Practice reverse calculations

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Written tests - Practical tasks

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 5 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 2	Lesson: 5
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.3: Scale Drawing - Interpreting linear scales in statement form

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Define linear scale - Interpret scale markings and express in statement form - Show understanding of scale representation

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do linear scales differ from numerical scales?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Maps with linear scales - Ruler - Pencil - Sample plans

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Examine linear scales on maps and plans - Measure length of scale markings - Determine what distance each unit represents - Practice with different linear scales - Express linear scales using words

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Oral questions - Written assignments

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 6 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 3	Lesson: 1
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.3: Scale Drawing - Writing linear scales in statement form

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Recall the format for writing scales in statement form - Express scales in statement form clearly and accurately - Demonstrate understanding of scale notation

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

Why is statement form useful for describing scales?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Ruler - Linear scale examples - Pencil - Drawing paper

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Express given scales in statement form - Write statements using proper format - Practice with scales showing various divisions - Convert linear scales to statements - Discuss advantages of statement form

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Written tests - Practical tasks

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 7 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 3	Lesson: 2
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.3: Scale Drawing - Interpreting linear scales in ratio form

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Define ratio form of scales - Convert measurement s to same units and express scales as ratios - Show understanding of proportional relationships

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

What does a scale ratio tell us about a drawing?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Ruler - Calculator - Conversion charts - Pencil

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Convert scales ensuring same units - Express scales as ratios - Practice unit conversions before writing ratios - Work with various scales - Understand ratios have no units

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Problem- solving - Oral questions

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 8 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 3	Lesson: 3
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.3: Scale Drawing - Interpreting linear scales in ratio form

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Define ratio form of scales - Convert measurement s to same units and express scales as ratios - Show understanding of proportional relationships

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

What does a scale ratio tell us about a drawing?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Ruler - Calculator - Conversion charts - Pencil

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Convert scales ensuring same units - Express scales as ratios - Practice unit conversions before writing ratios - Work with various scales - Understand ratios have no units

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Problem- solving - Oral questions

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 9 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 3	Lesson: 4
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.3: Scale Drawing - Writing linear scales in ratio form

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - State the requirements for writing scales in ratio form - Write scales in ratio form correctly without units - Demonstrate accuracy in conversions

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we ensure accuracy when converting to ratio form?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Calculator - Conversion tables - Pencil - Practice worksheets

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Complete tables converting statement to ratio form - Convert scales with various measurement s - Write map scales in ratio form - Calculate ratios for different scenarios - Practice systematic conversions

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Written assignments - Problem- solving

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 10 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 3	Lesson: 5
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.3: Scale Drawing - Converting scale from statement to ratio form

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - List the steps for converting statement to ratio form - Convert statement form scales to ratio form systematically - Show computational proficiency

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

What steps ensure correct conversion from statement to ratio?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Calculator - Ruler - Unit conversion chart - Pencil

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Convert statement scales to ratio form - Practice with different unit combinations - Apply systematic conversion process - Work with plans and maps - Verify conversions

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Written tests - Practical tasks

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 11 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 4	Lesson: 1
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.3: Scale Drawing - Converting scale from ratio to statement form

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain the process of converting ratio to statement form - Convert ratio form scales to statement form using appropriate units - Demonstrate understanding of both forms

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we choose appropriate units in statement form?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Atlas - Calculator - Ruler - Pencil

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Convert ratio scales to statement form - Determine appropriate units for actual measurement s - Express scales clearly in words - Practice with various ratio scales - Choose suitable units for statements

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Problem- solving - Oral questions

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 12 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 4	Lesson: 2
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.3: Scale Drawing - Converting scale from ratio to statement form

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain the process of converting ratio to statement form - Convert ratio form scales to statement form using appropriate units - Demonstrate understanding of both forms

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we choose appropriate units in statement form?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Atlas - Calculator - Ruler - Pencil

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Convert ratio scales to statement form - Determine appropriate units for actual measurement s - Express scales clearly in words - Practice with various ratio scales - Choose suitable units for statements

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Problem- solving - Oral questions

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 13 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 4	Lesson: 3
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.3: Scale Drawing - Making scale drawings with calculations

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Identify dimensions needed for scale drawings - Calculate scale lengths and make accurate scale drawings - Show precision in measurement s and drawing

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

Why must we calculate scale lengths before drawing?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Ruler - Pencil - Calculator - Drawing paper

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Calculate scale lengths before drawing - Make accurate scale drawings of various shapes - Apply appropriate scales - Measure and verify dimensions - Calculate areas from scale drawings

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Practical construction - Written tests

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 14 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 4	Lesson: 4
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.3: Scale Drawing - Scale drawings with distance calculations

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Recall how to measure distances on drawings - Make scale drawings involving multiple distances and calculate actual distances - Show systematic approach to problem- solving

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do scale drawings help solve distance problems?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Ruler - Pair of compasses - Calculator - Graph paper

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Make scale drawings involving multiple points - Use suitable scales for given distances - Measure lengths on scale drawings - Calculate actual distances from drawings - Apply geometric principles where needed - Verify measurement s

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Practical tasks - Problem- solving

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 15 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 4	Lesson: 5
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.3: Scale Drawing - Using maps and demonstrating scale

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Identify scales on actual maps - Read scales from maps and measure distances accurately - Appreciate real-world applications of scale

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How does scale choice affect what we can show on a map?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Atlas - Maps - Ruler - Calculator - Digital resources

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Examine maps in atlas - Identify and read map scales - Measure distances between locations - Calculate actual distances using scale - Compare different maps with different scales - Discuss map features

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Practical measurement - Oral questions

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 16 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 5	Lesson: 1
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.3: Scale Drawing - Application problems with scale

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Identify given information in scale problems - Solve complex problems involving scale, area, volume, time and speed - Show advanced problem- solving skills

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do professionals use scale in their work?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Calculator - Ruler - Problem cards - Reference materials

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Solve problems involving height and scale - Find scales used in given scenarios - Calculate areas from scale diagrams - Determine time and speed using map scales - Work with various measurement scenarios - Apply multiple concepts together

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Problem- solving - Written tests

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 17 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 5	Lesson: 2
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.3: Scale Drawing - Application problems with scale

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Identify given information in scale problems - Solve complex problems involving scale, area, volume, time and speed - Show advanced problem- solving skills

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do professionals use scale in their work?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Calculator - Ruler - Problem cards - Reference materials

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Solve problems involving height and scale - Find scales used in given scenarios - Calculate areas from scale diagrams - Determine time and speed using map scales - Work with various measurement scenarios - Apply multiple concepts together

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Problem- solving - Written tests

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 18 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 5	Lesson: 3
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.3: Scale Drawing - Using ICT for scale and maps

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Describe how digital maps use scale - Use digital devices to display maps and demonstrate zoom functions - Show digital literacy in geography context

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How does zooming affect the scale of a digital map?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Digital devices (tablets/comp uters) - Internet access - Digital mapping software - Projector

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Access digital maps on devices - Use zoom function to change scale - Observe how scale changes with zoom level - Measure distances on digital maps - Compare scale indicators on digital and paper maps - Discuss advantages of digital tools

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Practical demonstration - Oral questions

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 19 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 5	Lesson: 4
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.4: Common Solids - Identifying common solids from environment

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Name common solids: cubes, cuboids, cylinders, pyramids and cones - Classify solids by their properties - Show awareness of geometric shapes in environment

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

Where do we see these solids in our daily lives?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Collection of solid objects - Models of solids - Pictures of buildings - Digital images

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Collect objects from environment - Group objects by shape categories - Identify properties of each solid type - Discuss examples in daily life - Create display of classified solids

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Practical classification - Oral questions

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 20 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 5	Lesson: 5
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.4: Common Solids - Properties of solids (faces, edges, vertices)

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Define faces, edges and vertices - Identify and count faces, edges and vertices of given solids - Show understanding of 3D properties

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do faces, edges and vertices define a solid?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Models of solids - Ruler - Labels - Worksheet

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Examine labeled solids - Name all faces of solids - Identify all edges - Locate all vertices - Practice with different solids - Record properties systematically

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Written assignments - Practical identification

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 21 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 6	Lesson: 1
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.4: Common Solids - Sketching nets of cubes

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Define the term "net" of a solid - Sketch nets of closed and open cubes - Demonstrate spatial visualization

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How does a 3D cube transform into a 2D net?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model cubes - Scissors/razor blade - Ruler - Pencil - Plain paper

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Label cube vertices - Cut cube along specified edges - Lay out faces on flat surface - Sketch net showing all faces for closed cube - Sketch net showing appropriate faces for open cube - Identify different possible net arrangements

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Practical construction - Peer assessment

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 22 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 6	Lesson: 2
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.4: Common Solids - Sketching nets of cubes

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Define the term "net" of a solid - Sketch nets of closed and open cubes - Demonstrate spatial visualization

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How does a 3D cube transform into a 2D net?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model cubes - Scissors/razor blade - Ruler - Pencil - Plain paper

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Label cube vertices - Cut cube along specified edges - Lay out faces on flat surface - Sketch net showing all faces for closed cube - Sketch net showing appropriate faces for open cube - Identify different possible net arrangements

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Practical construction - Peer assessment

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 23 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 6	Lesson: 3
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.4: Common Solids - Sketching nets of cuboids

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Identify faces of cuboids - Sketch nets of closed and open cuboids - Show accuracy in net construction

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How does the net of a cuboid differ from that of a cube?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model cuboids - Scissors/razor blade - Ruler - Pencil - Grid paper

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Label cuboid vertices - Cut along specified edges - Spread faces on flat surface - Sketch net with all faces for closed cuboid - Sketch net with appropriate faces for open cuboid - Identify pairs of equal faces

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Practical tasks - Written tests

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 24 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 6	Lesson: 4
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.4: Common Solids - Sketching nets of cylinders

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Identify components of cylinder nets - Sketch nets of closed and open cylinders - Understand curved surface as rectangle - Show understanding of cylinder properties

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

Why does the curved surface become a rectangle?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model cylinders - Scissors/razor blade - Ruler - Pair of compasses - Pencil

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Cut cylinder to remove bases - Cut along height to open curved surface - Observe curved surface forms rectangle - Note rectangle dimensions relate to circumference - Sketch nets for closed and open cylinders - Label components

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Practical construction - Oral questions

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 25 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 6	Lesson: 5
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.4: Common Solids - Sketching nets of pyramids

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Describe components of pyramid nets - Sketch nets of pyramids with different bases - Show precision in drawing nets

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How many triangular faces does a square-based pyramid have?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model pyramids - Scissors/razor blade - Ruler - Pencil - Drawing paper

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Label pyramid vertices - Cut along slant edges - Lay faces on flat surface - Sketch net showing base and triangular faces - Ensure triangular faces connect to base edges - Practice with different base dimensions

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Practical tasks - Peer review

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 26 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 7	Lesson: 1
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.4: Common Solids - Sketching nets of cones

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Identify components of cone nets - Sketch nets of cones showing sector shape - Appreciate relationship between arc and circumference

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

Why does the cone's curved surface form a sector?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model cones - Scissors/razor blade - Protractor - Pair of compasses - Pencil

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Cut base from cone - Cut curved surface along slant height - Observe curved surface forms sector - Note relationship between arc length and base circumference - Sketch net showing circle and sector - Label components

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Practical construction - Written assignments

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 27 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 7	Lesson: 2
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.4: Common Solids - Sketching nets of cones

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Identify components of cone nets - Sketch nets of cones showing sector shape - Appreciate relationship between arc and circumference

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

Why does the cone's curved surface form a sector?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model cones - Scissors/razor blade - Protractor - Pair of compasses - Pencil

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Cut base from cone - Cut curved surface along slant height - Observe curved surface forms sector - Note relationship between arc length and base circumference - Sketch net showing circle and sector - Label components

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Practical construction - Written assignments

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 28 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 7	Lesson: 3
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.4: Common Solids - Matching solids to nets and vice versa

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Identify solids from their nets - Match given solids to correct nets - Demonstrate spatial reasoning

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How can we visualize the solid from looking at its net?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Various nets - Model solids - Ruler - Pencil - Matching cards

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Match various solids to their nets - Identify which solid each net will form - Draw solid that corresponds to given net - Practice visualizing 3D from 2D - Sketch nets for solids with various dimensions

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Practical matching - Problem- solving

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 29 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 7	Lesson: 4
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.4: Common Solids - Surface area of cubes from nets

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - State the formula for surface area of cube - Calculate total surface area of cube from its net - Show systematic calculation approach

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How does knowing one side help find total surface area?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model cubes - Ruler - Calculator - Pencil - Net templates

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Measure sides of cube - Sketch net of cube - Calculate area of one face - Multiply by number of faces - Practice with cubes of different dimensions - Verify by drawing net and calculating

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Written tests - Problem- solving

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 30 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 7	Lesson: 5
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.4: Common Solids - Surface area of cuboids from nets

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - State the formula for surface area of cuboid - Calculate total surface area of cuboid from nets - Show organized calculation method

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

Why do we calculate surface area in pairs for cuboids?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model cuboids - Ruler - Calculator - Grid paper - Pencil

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Draw net of cuboid with given dimensions - Calculate areas of different faces - Identify pairs of equal faces - Add all areas to find total - Practice with various dimensions - Verify calculations

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Written assignments - Practical calculations

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 31 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 8	Lesson: 1
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.4: Common Solids - Surface area of cylinders from nets

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - State components of cylinder surface area - Calculate total surface area of cylinder from nets - Demonstrate formula application

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How is the circumference used in finding surface area?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model cylinders - Ruler - Calculator - Pair of compasses - Formula chart

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Identify net components - Calculate area of circular faces - Find rectangle dimensions using circumference - Calculate rectangular area - Add areas for total surface area - Practice with different dimensions

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Problem- solving - Written tests

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 32 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 8	Lesson: 2
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.4: Common Solids - Surface area of cylinders from nets

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - State components of cylinder surface area - Calculate total surface area of cylinder from nets - Demonstrate formula application

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How is the circumference used in finding surface area?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model cylinders - Ruler - Calculator - Pair of compasses - Formula chart

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Identify net components - Calculate area of circular faces - Find rectangle dimensions using circumference - Calculate rectangular area - Add areas for total surface area - Practice with different dimensions

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Problem- solving - Written tests

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 33 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 8	Lesson: 3
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.4: Common Solids - Surface area of pyramids from nets

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Identify components of pyramid surface area - Calculate total surface area of pyramid from nets - Show systematic approach to complex calculations

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we find the slant height if not given?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model pyramids - Ruler - Calculator - Pencil - Net templates

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Draw net showing base and triangular faces - Calculate base area - Calculate area of each triangular face - Add base area to sum of triangular areas - Practice with different dimensions - Verify calculations

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Written assignments - Problem- solving

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 34 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 8	Lesson: 4
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.4: Common Solids - Surface area of cones and distance on surfaces

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - State formula for surface area of cone - Calculate surface area of cone from net and determine shortest distances on solid surfaces - Show advanced spatial reasoning

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How does opening a solid help find distances on its surface?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model cones and cuboids - Protractor - Calculator - String - Scissors

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Identify cone net components - Calculate circular base area - Calculate sector area using given angle - Find total surface area - Open cuboid into net to find paths between points - Measure distances along net surface

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Problem- solving - Practical tasks

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 35 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 8	Lesson: 5
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.4: Common Solids - Surface area of cones and distance on surfaces

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - State formula for surface area of cone - Calculate surface area of cone from net and determine shortest distances on solid surfaces - Show advanced spatial reasoning

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How does opening a solid help find distances on its surface?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model cones and cuboids - Protractor - Calculator - String - Scissors

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Identify cone net components - Calculate circular base area - Calculate sector area using given angle - Find total surface area - Open cuboid into net to find paths between points - Measure distances along net surface

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Problem- solving - Practical tasks

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 36 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 9	Lesson: 1
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.4: Common Solids - Making models of hollow solids (cubes and cuboids)

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - List steps for making hollow models - Construct hollow cube and cuboid models from nets - Show craftsmanship in model making

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

Why do we need flaps when making hollow models?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Manila paper - Ruler - Pencil - Scissors - Glue/paste - Colored markers

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Draw nets accurately on manila paper - Include flaps for joining faces - Cut out nets carefully - Fold along marked lines - Paste flaps to form hollow solids - Display completed models

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Practical construction - Peer assessment

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 37 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 9	Lesson: 2
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.4: Common Solids - Making models of cylinders, cones and pyramids

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain steps for making different hollow models - Construct hollow cylinder, cone and pyramid models - Show precision and craftsmanship

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we ensure the cylinder's rectangle matches the circle?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Manila paper - Pair of compasses - Protractor - Ruler - Scissors - Glue

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Draw cylinder net with calculated dimensions - Construct cone net with appropriate sector angle - Draw pyramid net with correct measurement s - Cut and fold to form solids - Paste edges to complete models - Display and compare models

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Practical construction - Written reflection

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 38 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 9	Lesson: 3
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.4: Common Solids - Using IT devices and drawing technology

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Identify technology tools for learning about solids - Use technology to explore and draw solids and nets - Appreciate technology in mathematics learning

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How does technology enhance our understanding of 3D shapes?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Computers/ta blets - Internet access - GeoGebra software - Projector - 3D modeling apps

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Watch educational videos about solids - Use software to draw 3D shapes - Explore rotating solids digitally - Practice drawing nets using technology - Use apps to visualize net folding - Share digital creations

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Digital portfolio - Oral presentation - Peer evaluation

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 39 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 9	Lesson: 4
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 5.0: Data Handling and Probability
Sub-strand / lesson focus: 5.1: Data Presentation and Interpretation - Collecting data and drawing bar graphs

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Define bar graph and identify its components - Collect data from own experiences and draw bar graphs with suitable scale - Appreciate the use of graphs in presenting data

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How can we represent collected data visually?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 197 - Ruler - Graph paper - Pencil - Data collection sheets

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Collect data from class members on given characteristics - Fill data in tables - Choose suitable scale for collected data - Draw bar graphs to represent collected data - Compare graphs with other groups - Discuss components of bar graphs

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Practical tasks - Oral questions

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 40 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 9	Lesson: 5
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 5.0: Data Handling and Probability
Sub-strand / lesson focus: 5.1: Data Presentation and Interpretation - Drawing bar graphs with suitable scale 5.1: Data Presentation and Interpretation - Interpreting bar graphs

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - State the steps for drawing bar graphs - Draw bar graphs with appropriate scales for different data sets - Show accuracy in graph construction

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we choose an appropriate scale for a bar graph?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 197 - Graph paper - Ruler - Pencil - Calculator - Data tables - Sample bar graphs - Question sheets

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Choose uniform width for bars - Select uniform gaps between bars - Choose suitable scale for vertical axis - Calculate heights of bars according to scale - Draw bars accurately - Label axes properly - Practice with various data sets

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Practical construction - Written assignments

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 41 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 10	Lesson: 1
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 5.0: Data Handling and Probability
Sub-strand / lesson focus: 5.1: Data Presentation and Interpretation - Drawing line graphs

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Define line graph and state its uses - Draw line graphs from given data - Appreciate line graphs for showing trends

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

When is it appropriate to use a line graph instead of a bar graph?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 197 - Graph paper - Ruler - Pencil - Calculator - Data tables

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Choose suitable scale for x-axis - Choose suitable scale for y-axis - Plot points from table of values - Join plotted points using straight lines - Label axes appropriately - Practice drawing line graphs for different data sets

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Practical construction - Peer assessment

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 42 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 10	Lesson: 2
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 5.0: Data Handling and Probability
Sub-strand / lesson focus: 5.1: Data Presentation and Interpretation - Interpreting line graphs 5.1: Data Presentation and Interpretation - Identifying mode of discrete data

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain how to read values from line graphs - Interpret line graphs to extract information - Show analytical skills in reading trends

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do line graphs help us see changes over time?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 197 - Sample line graphs - Ruler - Pencil - Question sheets - Number cards - Exercise books - Data sets

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Read values at specific points on graph - Identify highest and lowest values - Determine trends from line graphs - Calculate totals from graph data - Answer questions based on line graphs - Discuss patterns observed

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Written tests - Problem- solving

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 43 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 10	Lesson: 3
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 5.0: Data Handling and Probability
Sub-strand / lesson focus: 5.1: Data Presentation and Interpretation - Calculating mean of discrete data 5.1: Data Presentation and Interpretation - Working out averages from different sets

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - State the formula for calculating mean - Calculate the mean of discrete data sets accurately - Show systematic approach in calculations

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How does the mean represent the average of a data set?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 197 - Calculator - Pencil - Exercise books - Data sets - Problem cards

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Note down values from group members - Add all values in data set - Count number of values - Divide sum by number of values - Calculate mean for various data sets - Verify calculations - Practice with different contexts

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Written tests - Problem- solving

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 44 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 10	Lesson: 4
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 5.0: Data Handling and Probability
Sub-strand / lesson focus: 5.1: Data Presentation and Interpretation - Determining median of discrete data

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Define median and explain the process of finding it - Determine the median of discrete data for odd and even sets - Show systematic approach in finding median

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

Why must data be arranged in order before finding the median?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 197 - Number cards - Pencil - Exercise books - Calculator

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Arrange data in ascending or descending order - Identify middle value for odd sets - Calculate median for even sets by averaging two middle values - Practice finding median for various data sets - Compare median with mode and mean - Discuss applications

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Oral questions - Written tests

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 45 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 10	Lesson: 5
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 5.0: Data Handling and Probability
Sub-strand / lesson focus: 5.1: Data Presentation and Interpretation - Using IT for data presentation and calculations

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Identify IT tools for creating graphs - Use technology to create bar graphs and line graphs and calculate mean, mode and median - Appreciate technology in data handling

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How does technology make data presentation and analysis easier?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 197 - Computers/ta blets - Spreadsheet software - Internet access - Projector - Data sets

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Use spreadsheet software to enter data - Create bar graphs using software - Create line graphs using software - Use formulas to calculate mean - Use functions to find mode and median - Compare manual and digital methods - Present findings digitally

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Digital portfolio - Practical demonstration - Peer evaluation

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 46 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 11	Lesson: 1
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 5.0: Data Handling and Probability
Sub-strand / lesson focus: 5.2: Probability - Identifying events involving chance in real life

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Define chance and probability - Identify events involving chance in daily life - Show awareness of probability in real situations

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

What is chance and where do we encounter it in daily life?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 210 - Pictures of chance events - Pencil - Chart paper - Real- life scenario cards

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Discuss possibilities in various scenarios - Identify chance events in sports - Recognize chance in weather predictions - Discuss chance in games - List daily events involving chance - Share observations with class

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Oral questions - Class discussion

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 47 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 11	Lesson: 2
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 5.0: Data Handling and Probability
Sub-strand / lesson focus: 5.2: Probability - Identifying events involving chance in real life

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Define chance and probability - Identify events involving chance in daily life - Show awareness of probability in real situations

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

What is chance and where do we encounter it in daily life?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 210 - Pictures of chance events - Pencil - Chart paper - Real- life scenario cards

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Discuss possibilities in various scenarios - Identify chance events in sports - Recognize chance in weather predictions - Discuss chance in games - List daily events involving chance - Share observations with class

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Oral questions - Class discussion

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 48 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 11	Lesson: 3
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 5.0: Data Handling and Probability
Sub-strand / lesson focus: 5.2: Probability - Discussing likely and unlikely events

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - List the likelihood scale terms: impossible, unlikely, equally likely, likely, certain - Classify events as impossible, unlikely, equally likely, likely or certain - Show critical thinking in analyzing probability

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we describe the likelihood of different events happening?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 210 - Likelihood scale chart - Event cards - Pencil - Exercise books

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Examine likelihood scale - Discuss meaning of each term - Classify statements using likelihood terms - Identify impossible events - Identify certain events - Distinguish between likely and unlikely - Practice with various statements

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Oral questions - Written assignments

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 49 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 11	Lesson: 4
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 5.0: Data Handling and Probability
Sub-strand / lesson focus: 5.2: Probability - Performing chance experiments

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Define chance experiment - Perform chance experiments such as flipping coins, tossing dice, and drawing objects - Show interest in hands-on probability activities

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

What are the possible outcomes when we perform chance experiments?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 210 - Coins - Dice - Colored balls/beads - Bags - Spinners - Recording sheets

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Obtain coins and flip them - Toss dice and record outcomes - Draw colored balls or beads from bags - Use spinners and record results - Record outcomes from experiments - Compare results with other groups - Discuss patterns observed

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Practical tasks - Oral questions

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 50 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 11	Lesson: 5
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 5.0: Data Handling and
Sub-strand / lesson focus: 5.2: Probability - Writing experimental probability outcomes

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain the concept of experimental probability - Write all possible outcomes from chance experiments - Demonstrate systematic recording of outcomes

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we list all possible outcomes from an experiment?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 210 - Coins - Dice - Number cards - Pencil - Exercise books

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - List possible outcomes from coin toss - Write outcomes from die roll - Determine outcomes from spinners - List outcomes from drawing objects - Form combinations of outcomes - Record outcomes systematically - Share findings with class

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Written tests - Problem- solving

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 51 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 12	Lesson: 1
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 5.0: Data Handling and Probability
Sub-strand / lesson focus: 5.2: Probability - Expressing probability outcomes as fractions

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - State the formula for probability as a fraction - Express probability outcomes as fractions accurately - Show understanding of favorable outcomes

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we express the chance of an event happening as a fraction?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 210 - Colored balls/beads - Bags - Calculator - Pencil - Exercise books

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Identify total possible outcomes - Identify favorable outcomes - Express probability as fraction of favorable to total outcomes - Simplify probability fractions - Calculate probabilities from various scenarios - Solve word problems involving probability - Verify answers

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Written assignments - Problem- solving tasks

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 52 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 12	Lesson: 2
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 5.0: Data Handling and Probability
Sub-strand / lesson focus: 5.2: Probability - Expressing probability outcomes as fractions

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - State the formula for probability as a fraction - Express probability outcomes as fractions accurately - Show understanding of favorable outcomes

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we express the chance of an event happening as a fraction?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 210 - Colored balls/beads - Bags - Calculator - Pencil - Exercise books

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Identify total possible outcomes - Identify favorable outcomes - Express probability as fraction of favorable to total outcomes - Simplify probability fractions - Calculate probabilities from various scenarios - Solve word problems involving probability - Verify answers

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Written assignments - Problem- solving tasks

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 53 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 12	Lesson: 3
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 5.0: Data Handling and Probability
Sub-strand / lesson focus: 5.2: Probability - Expressing probability as decimals and percentages

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain the relationship between probability in fractions, decimals and percentages - Convert probability from fractions to decimals and percentages - Demonstrate proficiency in probability conversions

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

Why is probability sometimes expressed as decimals or percentages?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 210 - Calculator - Pencil - Exercise books - Conversion charts

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Convert probability fractions to decimals - Convert probability fractions to percentages - Understand that probability in decimals cannot exceed 1 - Understand that probability in percentages cannot exceed 100% - Calculate complementar y probabilities - Solve problems in different forms - Apply probability in real contexts

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Written tests - Problem- solving

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 54 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 12	Lesson: 4
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 5.0: Data Handling and Probability
Sub-strand / lesson focus: 5.2: Probability - Using IT to play probability games

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Identify digital tools for probability activities - Use technology to play games involving probability and simulate experiments - Appreciate technology in learning probability

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How does technology help us understand probability better?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 210 - Computers/ta blets - Internet access - Probability apps/software - Projector - Recording sheets

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Access online probability games - Use software to simulate coin flips - Use apps to simulate dice rolls - Play digital probability games - Record results from digital experiments - Compare manual and digital experiments - Discuss advantages of using technology

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Digital portfolio - Practical demonstration - Oral presentation

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 2

LESSON PLAN 55 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 12	Lesson: 5
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 5.0: Data Handling and Probability
Sub-strand / lesson focus: 5.2: Probability - Using IT to play probability games

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to:

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

LEARNING RESOURCES

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to:

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 1 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 2	Lesson: 1
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.3: Scale Drawing - Representatio n of length to given scale

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Define scale and its purpose - Determine scale from given measurement s - Show understanding of proportion

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

Why do we need scale when drawing large objects?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Ruler - Tape measure - Pencil - Drawing paper

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Compare sizes of objects and their representation s - Discuss need for scale in drawings - Measure actual dimensions - Choose appropriate scale for representation s - Calculate scale from given information - Express scale in different forms

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Oral questions - Practical tasks

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 2 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 2	Lesson: 2
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.3: Scale Drawing - Converting actual length to scale length

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - State the formula for converting actual length to scale length - Convert actual measurement s to scale measurement s accurately - Demonstrate computational skills

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we calculate scale length from actual length?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Calculator - Ruler - Conversion tables - Pencil

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Apply given scales to convert measurement s - Complete tables converting actual to scale lengths - Calculate scale lengths using various scales - Work with different units - Practice systematic conversions

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Written assignments - Problem- solving

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 3 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 2	Lesson: 3
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.3: Scale Drawing - Converting actual length to scale length

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - State the formula for converting actual length to scale length - Convert actual measurement s to scale measurement s accurately - Demonstrate computational skills

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we calculate scale length from actual length?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Calculator - Ruler - Conversion tables - Pencil

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Apply given scales to convert measurement s - Complete tables converting actual to scale lengths - Calculate scale lengths using various scales - Work with different units - Practice systematic conversions

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Written assignments - Problem- solving

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 4 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 2	Lesson: 4
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.3: Scale Drawing - Converting scale length to actual length

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain the process of converting scale to actual measurement s - Convert scale measurement s to actual measurement s accurately - Show systematic calculation approach

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we find real dimensions from scale drawings?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Ruler - Calculator - Scale drawings - Pencil

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Measure lengths on scale diagrams - Use given scales to find actual lengths - Calculate actual distances - Work with different unit conversions - Practice reverse calculations

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Written tests - Practical tasks

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 5 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 2	Lesson: 5
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.3: Scale Drawing - Interpreting linear scales in statement form

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Define linear scale - Interpret scale markings and express in statement form - Show understanding of scale representation

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do linear scales differ from numerical scales?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Maps with linear scales - Ruler - Pencil - Sample plans

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Examine linear scales on maps and plans - Measure length of scale markings - Determine what distance each unit represents - Practice with different linear scales - Express linear scales using words

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Oral questions - Written assignments

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 6 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 3	Lesson: 1
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.3: Scale Drawing - Writing linear scales in statement form

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Recall the format for writing scales in statement form - Express scales in statement form clearly and accurately - Demonstrate understanding of scale notation

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

Why is statement form useful for describing scales?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Ruler - Linear scale examples - Pencil - Drawing paper

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Express given scales in statement form - Write statements using proper format - Practice with scales showing various divisions - Convert linear scales to statements - Discuss advantages of statement form

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Written tests - Practical tasks

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 7 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 3	Lesson: 2
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.3: Scale Drawing - Interpreting linear scales in ratio form

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Define ratio form of scales - Convert measurement s to same units and express scales as ratios - Show understanding of proportional relationships

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

What does a scale ratio tell us about a drawing?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Ruler - Calculator - Conversion charts - Pencil

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Convert scales ensuring same units - Express scales as ratios - Practice unit conversions before writing ratios - Work with various scales - Understand ratios have no units

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Problem- solving - Oral questions

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 8 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 3	Lesson: 3
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.3: Scale Drawing - Interpreting linear scales in ratio form

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Define ratio form of scales - Convert measurement s to same units and express scales as ratios - Show understanding of proportional relationships

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

What does a scale ratio tell us about a drawing?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Ruler - Calculator - Conversion charts - Pencil

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Convert scales ensuring same units - Express scales as ratios - Practice unit conversions before writing ratios - Work with various scales - Understand ratios have no units

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Problem- solving - Oral questions

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 9 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 3	Lesson: 4
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.3: Scale Drawing - Writing linear scales in ratio form

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - State the requirements for writing scales in ratio form - Write scales in ratio form correctly without units - Demonstrate accuracy in conversions

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we ensure accuracy when converting to ratio form?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Calculator - Conversion tables - Pencil - Practice worksheets

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Complete tables converting statement to ratio form - Convert scales with various measurement s - Write map scales in ratio form - Calculate ratios for different scenarios - Practice systematic conversions

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Written assignments - Problem- solving

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 10 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 3	Lesson: 5
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.3: Scale Drawing - Converting scale from statement to ratio form

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - List the steps for converting statement to ratio form - Convert statement form scales to ratio form systematically - Show computational proficiency

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

What steps ensure correct conversion from statement to ratio?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Calculator - Ruler - Unit conversion chart - Pencil

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Convert statement scales to ratio form - Practice with different unit combinations - Apply systematic conversion process - Work with plans and maps - Verify conversions

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Written tests - Practical tasks

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 11 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 4	Lesson: 1
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.3: Scale Drawing - Converting scale from ratio to statement form

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain the process of converting ratio to statement form - Convert ratio form scales to statement form using appropriate units - Demonstrate understanding of both forms

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we choose appropriate units in statement form?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Atlas - Calculator - Ruler - Pencil

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Convert ratio scales to statement form - Determine appropriate units for actual measurement s - Express scales clearly in words - Practice with various ratio scales - Choose suitable units for statements

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Problem- solving - Oral questions

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 12 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 4	Lesson: 2
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.3: Scale Drawing - Converting scale from ratio to statement form

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain the process of converting ratio to statement form - Convert ratio form scales to statement form using appropriate units - Demonstrate understanding of both forms

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we choose appropriate units in statement form?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Atlas - Calculator - Ruler - Pencil

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Convert ratio scales to statement form - Determine appropriate units for actual measurement s - Express scales clearly in words - Practice with various ratio scales - Choose suitable units for statements

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Problem- solving - Oral questions

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 13 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 4	Lesson: 3
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.3: Scale Drawing - Making scale drawings with calculations

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Identify dimensions needed for scale drawings - Calculate scale lengths and make accurate scale drawings - Show precision in measurement s and drawing

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

Why must we calculate scale lengths before drawing?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Ruler - Pencil - Calculator - Drawing paper

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Calculate scale lengths before drawing - Make accurate scale drawings of various shapes - Apply appropriate scales - Measure and verify dimensions - Calculate areas from scale drawings

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Practical construction - Written tests

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 14 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 4	Lesson: 4
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.3: Scale Drawing - Scale drawings with distance calculations

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Recall how to measure distances on drawings - Make scale drawings involving multiple distances and calculate actual distances - Show systematic approach to problem- solving

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do scale drawings help solve distance problems?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Ruler - Pair of compasses - Calculator - Graph paper

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Make scale drawings involving multiple points - Use suitable scales for given distances - Measure lengths on scale drawings - Calculate actual distances from drawings - Apply geometric principles where needed - Verify measurement s

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Practical tasks - Problem- solving

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 15 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 4	Lesson: 5
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.3: Scale Drawing - Using maps and demonstrating scale

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Identify scales on actual maps - Read scales from maps and measure distances accurately - Appreciate real-world applications of scale

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How does scale choice affect what we can show on a map?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Atlas - Maps - Ruler - Calculator - Digital resources

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Examine maps in atlas - Identify and read map scales - Measure distances between locations - Calculate actual distances using scale - Compare different maps with different scales - Discuss map features

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Practical measurement - Oral questions

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 16 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 5	Lesson: 1
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.3: Scale Drawing - Application problems with scale

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Identify given information in scale problems - Solve complex problems involving scale, area, volume, time and speed - Show advanced problem- solving skills

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do professionals use scale in their work?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Calculator - Ruler - Problem cards - Reference materials

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Solve problems involving height and scale - Find scales used in given scenarios - Calculate areas from scale diagrams - Determine time and speed using map scales - Work with various measurement scenarios - Apply multiple concepts together

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Problem- solving - Written tests

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 17 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 5	Lesson: 2
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.3: Scale Drawing - Application problems with scale

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Identify given information in scale problems - Solve complex problems involving scale, area, volume, time and speed - Show advanced problem- solving skills

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do professionals use scale in their work?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Calculator - Ruler - Problem cards - Reference materials

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Solve problems involving height and scale - Find scales used in given scenarios - Calculate areas from scale diagrams - Determine time and speed using map scales - Work with various measurement scenarios - Apply multiple concepts together

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Problem- solving - Written tests

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 18 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 5	Lesson: 3
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.3: Scale Drawing - Using ICT for scale and maps

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Describe how digital maps use scale - Use digital devices to display maps and demonstrate zoom functions - Show digital literacy in geography context

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How does zooming affect the scale of a digital map?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 160 - Digital devices (tablets/comp uters) - Internet access - Digital mapping software - Projector

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Access digital maps on devices - Use zoom function to change scale - Observe how scale changes with zoom level - Measure distances on digital maps - Compare scale indicators on digital and paper maps - Discuss advantages of digital tools

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Practical demonstration - Oral questions

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 19 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 5	Lesson: 4
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.4: Common Solids - Identifying common solids from environment

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Name common solids: cubes, cuboids, cylinders, pyramids and cones - Classify solids by their properties - Show awareness of geometric shapes in environment

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

Where do we see these solids in our daily lives?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Collection of solid objects - Models of solids - Pictures of buildings - Digital images

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Collect objects from environment - Group objects by shape categories - Identify properties of each solid type - Discuss examples in daily life - Create display of classified solids

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Practical classification - Oral questions

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 20 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 5	Lesson: 5
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.4: Common Solids - Properties of solids (faces, edges, vertices)

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Define faces, edges and vertices - Identify and count faces, edges and vertices of given solids - Show understanding of 3D properties

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do faces, edges and vertices define a solid?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Models of solids - Ruler - Labels - Worksheet

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Examine labeled solids - Name all faces of solids - Identify all edges - Locate all vertices - Practice with different solids - Record properties systematically

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Written assignments - Practical identification

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 21 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 6	Lesson: 1
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.4: Common Solids - Sketching nets of cubes

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Define the term "net" of a solid - Sketch nets of closed and open cubes - Demonstrate spatial visualization

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How does a 3D cube transform into a 2D net?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model cubes - Scissors/razor blade - Ruler - Pencil - Plain paper

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Label cube vertices - Cut cube along specified edges - Lay out faces on flat surface - Sketch net showing all faces for closed cube - Sketch net showing appropriate faces for open cube - Identify different possible net arrangements

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Practical construction - Peer assessment

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 22 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 6	Lesson: 2
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.4: Common Solids - Sketching nets of cubes

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Define the term "net" of a solid - Sketch nets of closed and open cubes - Demonstrate spatial visualization

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How does a 3D cube transform into a 2D net?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model cubes - Scissors/razor blade - Ruler - Pencil - Plain paper

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Label cube vertices - Cut cube along specified edges - Lay out faces on flat surface - Sketch net showing all faces for closed cube - Sketch net showing appropriate faces for open cube - Identify different possible net arrangements

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Practical construction - Peer assessment

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 23 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 6	Lesson: 3
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.4: Common Solids - Sketching nets of cuboids

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Identify faces of cuboids - Sketch nets of closed and open cuboids - Show accuracy in net construction

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How does the net of a cuboid differ from that of a cube?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model cuboids - Scissors/razor blade - Ruler - Pencil - Grid paper

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Label cuboid vertices - Cut along specified edges - Spread faces on flat surface - Sketch net with all faces for closed cuboid - Sketch net with appropriate faces for open cuboid - Identify pairs of equal faces

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Practical tasks - Written tests

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 24 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 6	Lesson: 4
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.4: Common Solids - Sketching nets of cylinders

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Identify components of cylinder nets - Sketch nets of closed and open cylinders - Understand curved surface as rectangle - Show understanding of cylinder properties

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

Why does the curved surface become a rectangle?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model cylinders - Scissors/razor blade - Ruler - Pair of compasses - Pencil

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Cut cylinder to remove bases - Cut along height to open curved surface - Observe curved surface forms rectangle - Note rectangle dimensions relate to circumference - Sketch nets for closed and open cylinders - Label components

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Practical construction - Oral questions

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 25 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 6	Lesson: 5
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.4: Common Solids - Sketching nets of pyramids

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Describe components of pyramid nets - Sketch nets of pyramids with different bases - Show precision in drawing nets

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How many triangular faces does a square-based pyramid have?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model pyramids - Scissors/razor blade - Ruler - Pencil - Drawing paper

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Label pyramid vertices - Cut along slant edges - Lay faces on flat surface - Sketch net showing base and triangular faces - Ensure triangular faces connect to base edges - Practice with different base dimensions

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Practical tasks - Peer review

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 26 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 7	Lesson: 1
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.4: Common Solids - Sketching nets of cones

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Identify components of cone nets - Sketch nets of cones showing sector shape - Appreciate relationship between arc and circumference

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

Why does the cone's curved surface form a sector?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model cones - Scissors/razor blade - Protractor - Pair of compasses - Pencil

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Cut base from cone - Cut curved surface along slant height - Observe curved surface forms sector - Note relationship between arc length and base circumference - Sketch net showing circle and sector - Label components

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Practical construction - Written assignments

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 27 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 7	Lesson: 2
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.4: Common Solids - Sketching nets of cones

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Identify components of cone nets - Sketch nets of cones showing sector shape - Appreciate relationship between arc and circumference

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

Why does the cone's curved surface form a sector?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model cones - Scissors/razor blade - Protractor - Pair of compasses - Pencil

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Cut base from cone - Cut curved surface along slant height - Observe curved surface forms sector - Note relationship between arc length and base circumference - Sketch net showing circle and sector - Label components

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Practical construction - Written assignments

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 28 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 7	Lesson: 3
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.4: Common Solids - Matching solids to nets and vice versa

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Identify solids from their nets - Match given solids to correct nets - Demonstrate spatial reasoning

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How can we visualize the solid from looking at its net?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Various nets - Model solids - Ruler - Pencil - Matching cards

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Match various solids to their nets - Identify which solid each net will form - Draw solid that corresponds to given net - Practice visualizing 3D from 2D - Sketch nets for solids with various dimensions

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Practical matching - Problem- solving

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 29 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 7	Lesson: 4
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.4: Common Solids - Surface area of cubes from nets

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - State the formula for surface area of cube - Calculate total surface area of cube from its net - Show systematic calculation approach

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How does knowing one side help find total surface area?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model cubes - Ruler - Calculator - Pencil - Net templates

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Measure sides of cube - Sketch net of cube - Calculate area of one face - Multiply by number of faces - Practice with cubes of different dimensions - Verify by drawing net and calculating

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Written tests - Problem- solving

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 30 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 7	Lesson: 5
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.4: Common Solids - Surface area of cuboids from nets

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - State the formula for surface area of cuboid - Calculate total surface area of cuboid from nets - Show organized calculation method

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

Why do we calculate surface area in pairs for cuboids?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model cuboids - Ruler - Calculator - Grid paper - Pencil

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Draw net of cuboid with given dimensions - Calculate areas of different faces - Identify pairs of equal faces - Add all areas to find total - Practice with various dimensions - Verify calculations

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Written assignments - Practical calculations

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 31 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 8	Lesson: 1
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.4: Common Solids - Surface area of cylinders from nets

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - State components of cylinder surface area - Calculate total surface area of cylinder from nets - Demonstrate formula application

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How is the circumference used in finding surface area?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model cylinders - Ruler - Calculator - Pair of compasses - Formula chart

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Identify net components - Calculate area of circular faces - Find rectangle dimensions using circumference - Calculate rectangular area - Add areas for total surface area - Practice with different dimensions

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Problem- solving - Written tests

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 32 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 8	Lesson: 2
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.4: Common Solids - Surface area of cylinders from nets

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - State components of cylinder surface area - Calculate total surface area of cylinder from nets - Demonstrate formula application

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How is the circumference used in finding surface area?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model cylinders - Ruler - Calculator - Pair of compasses - Formula chart

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Identify net components - Calculate area of circular faces - Find rectangle dimensions using circumference - Calculate rectangular area - Add areas for total surface area - Practice with different dimensions

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Problem- solving - Written tests

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 33 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 8	Lesson: 3
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.4: Common Solids - Surface area of pyramids from nets

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Identify components of pyramid surface area - Calculate total surface area of pyramid from nets - Show systematic approach to complex calculations

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we find the slant height if not given?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model pyramids - Ruler - Calculator - Pencil - Net templates

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Draw net showing base and triangular faces - Calculate base area - Calculate area of each triangular face - Add base area to sum of triangular areas - Practice with different dimensions - Verify calculations

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Written assignments - Problem- solving

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 34 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 8	Lesson: 4
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.4: Common Solids - Surface area of cones and distance on surfaces

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - State formula for surface area of cone - Calculate surface area of cone from net and determine shortest distances on solid surfaces - Show advanced spatial reasoning

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How does opening a solid help find distances on its surface?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model cones and cuboids - Protractor - Calculator - String - Scissors

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Identify cone net components - Calculate circular base area - Calculate sector area using given angle - Find total surface area - Open cuboid into net to find paths between points - Measure distances along net surface

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Problem- solving - Practical tasks

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 35 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 8	Lesson: 5
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.4: Common Solids - Surface area of cones and distance on surfaces

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - State formula for surface area of cone - Calculate surface area of cone from net and determine shortest distances on solid surfaces - Show advanced spatial reasoning

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How does opening a solid help find distances on its surface?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Model cones and cuboids - Protractor - Calculator - String - Scissors

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Identify cone net components - Calculate circular base area - Calculate sector area using given angle - Find total surface area - Open cuboid into net to find paths between points - Measure distances along net surface

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Problem- solving - Practical tasks

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 36 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 9	Lesson: 1
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.4: Common Solids - Making models of hollow solids (cubes and cuboids)

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - List steps for making hollow models - Construct hollow cube and cuboid models from nets - Show craftsmanship in model making

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

Why do we need flaps when making hollow models?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Manila paper - Ruler - Pencil - Scissors - Glue/paste - Colored markers

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Draw nets accurately on manila paper - Include flaps for joining faces - Cut out nets carefully - Fold along marked lines - Paste flaps to form hollow solids - Display completed models

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Practical construction - Peer assessment

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 37 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 9	Lesson: 2
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.4: Common Solids - Making models of cylinders, cones and pyramids

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain steps for making different hollow models - Construct hollow cylinder, cone and pyramid models - Show precision and craftsmanship

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we ensure the cylinder's rectangle matches the circle?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Manila paper - Pair of compasses - Protractor - Ruler - Scissors - Glue

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Draw cylinder net with calculated dimensions - Construct cone net with appropriate sector angle - Draw pyramid net with correct measurement s - Cut and fold to form solids - Paste edges to complete models - Display and compare models

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Practical construction - Written reflection

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 38 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 9	Lesson: 3
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 4.0: Geometry
Sub-strand / lesson focus: 4.4: Common Solids - Using IT devices and drawing technology

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Identify technology tools for learning about solids - Use technology to explore and draw solids and nets - Appreciate technology in mathematics learning

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How does technology enhance our understanding of 3D shapes?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 176 - Computers/ta blets - Internet access - GeoGebra software - Projector - 3D modeling apps

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Watch educational videos about solids - Use software to draw 3D shapes - Explore rotating solids digitally - Practice drawing nets using technology - Use apps to visualize net folding - Share digital creations

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Digital portfolio - Oral presentation - Peer evaluation

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 39 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 9	Lesson: 4
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 5.0: Data Handling and Probability
Sub-strand / lesson focus: 5.1: Data Presentation and Interpretation - Collecting data and drawing bar graphs

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Define bar graph and identify its components - Collect data from own experiences and draw bar graphs with suitable scale - Appreciate the use of graphs in presenting data

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How can we represent collected data visually?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 197 - Ruler - Graph paper - Pencil - Data collection sheets

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Collect data from class members on given characteristics - Fill data in tables - Choose suitable scale for collected data - Draw bar graphs to represent collected data - Compare graphs with other groups - Discuss components of bar graphs

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Practical tasks - Oral questions

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 40 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 9	Lesson: 5
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 5.0: Data Handling and Probability
Sub-strand / lesson focus: 5.1: Data Presentation and Interpretation - Drawing bar graphs with suitable scale 5.1: Data Presentation and Interpretation - Interpreting bar graphs

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - State the steps for drawing bar graphs - Draw bar graphs with appropriate scales for different data sets - Show accuracy in graph construction

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we choose an appropriate scale for a bar graph?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 197 - Graph paper - Ruler - Pencil - Calculator - Data tables - Sample bar graphs - Question sheets

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Choose uniform width for bars - Select uniform gaps between bars - Choose suitable scale for vertical axis - Calculate heights of bars according to scale - Draw bars accurately - Label axes properly - Practice with various data sets

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Practical construction - Written assignments

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 41 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 10	Lesson: 1
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 5.0: Data Handling and Probability
Sub-strand / lesson focus: 5.1: Data Presentation and Interpretation - Drawing line graphs

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Define line graph and state its uses - Draw line graphs from given data - Appreciate line graphs for showing trends

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

When is it appropriate to use a line graph instead of a bar graph?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 197 - Graph paper - Ruler - Pencil - Calculator - Data tables

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Choose suitable scale for x-axis - Choose suitable scale for y-axis - Plot points from table of values - Join plotted points using straight lines - Label axes appropriately - Practice drawing line graphs for different data sets

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Practical construction - Peer assessment

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 42 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 10	Lesson: 2
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 5.0: Data Handling and Probability
Sub-strand / lesson focus: 5.1: Data Presentation and Interpretation - Interpreting line graphs 5.1: Data Presentation and Interpretation - Identifying mode of discrete data

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain how to read values from line graphs - Interpret line graphs to extract information - Show analytical skills in reading trends

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do line graphs help us see changes over time?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 197 - Sample line graphs - Ruler - Pencil - Question sheets - Number cards - Exercise books - Data sets

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Read values at specific points on graph - Identify highest and lowest values - Determine trends from line graphs - Calculate totals from graph data - Answer questions based on line graphs - Discuss patterns observed

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Written tests - Problem- solving

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 43 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 10	Lesson: 3
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 5.0: Data Handling and Probability
Sub-strand / lesson focus: 5.1: Data Presentation and Interpretation - Calculating mean of discrete data 5.1: Data Presentation and Interpretation - Working out averages from different sets

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - State the formula for calculating mean - Calculate the mean of discrete data sets accurately - Show systematic approach in calculations

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How does the mean represent the average of a data set?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 197 - Calculator - Pencil - Exercise books - Data sets - Problem cards

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Note down values from group members - Add all values in data set - Count number of values - Divide sum by number of values - Calculate mean for various data sets - Verify calculations - Practice with different contexts

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Written tests - Problem- solving

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 44 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 10	Lesson: 4
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 5.0: Data Handling and Probability
Sub-strand / lesson focus: 5.1: Data Presentation and Interpretation - Determining median of discrete data

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Define median and explain the process of finding it - Determine the median of discrete data for odd and even sets - Show systematic approach in finding median

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

Why must data be arranged in order before finding the median?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 197 - Number cards - Pencil - Exercise books - Calculator

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Arrange data in ascending or descending order - Identify middle value for odd sets - Calculate median for even sets by averaging two middle values - Practice finding median for various data sets - Compare median with mode and mean - Discuss applications

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Oral questions - Written tests

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 45 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 10	Lesson: 5
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 5.0: Data Handling and Probability
Sub-strand / lesson focus: 5.1: Data Presentation and Interpretation - Using IT for data presentation and calculations

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Identify IT tools for creating graphs - Use technology to create bar graphs and line graphs and calculate mean, mode and median - Appreciate technology in data handling

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How does technology make data presentation and analysis easier?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 197 - Computers/ta blets - Spreadsheet software - Internet access - Projector - Data sets

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Use spreadsheet software to enter data - Create bar graphs using software - Create line graphs using software - Use formulas to calculate mean - Use functions to find mode and median - Compare manual and digital methods - Present findings digitally

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Digital portfolio - Practical demonstration - Peer evaluation

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 46 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 11	Lesson: 1
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 5.0: Data Handling and Probability
Sub-strand / lesson focus: 5.2: Probability - Identifying events involving chance in real life

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Define chance and probability - Identify events involving chance in daily life - Show awareness of probability in real situations

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

What is chance and where do we encounter it in daily life?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 210 - Pictures of chance events - Pencil - Chart paper - Real- life scenario cards

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Discuss possibilities in various scenarios - Identify chance events in sports - Recognize chance in weather predictions - Discuss chance in games - List daily events involving chance - Share observations with class

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Oral questions - Class discussion

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 47 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 11	Lesson: 2
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 5.0: Data Handling and Probability
Sub-strand / lesson focus: 5.2: Probability - Identifying events involving chance in real life

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Define chance and probability - Identify events involving chance in daily life - Show awareness of probability in real situations

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

What is chance and where do we encounter it in daily life?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 210 - Pictures of chance events - Pencil - Chart paper - Real- life scenario cards

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Discuss possibilities in various scenarios - Identify chance events in sports - Recognize chance in weather predictions - Discuss chance in games - List daily events involving chance - Share observations with class

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Oral questions - Class discussion

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 48 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 11	Lesson: 3
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 5.0: Data Handling and Probability
Sub-strand / lesson focus: 5.2: Probability - Discussing likely and unlikely events

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - List the likelihood scale terms: impossible, unlikely, equally likely, likely, certain - Classify events as impossible, unlikely, equally likely, likely or certain - Show critical thinking in analyzing probability

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we describe the likelihood of different events happening?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 210 - Likelihood scale chart - Event cards - Pencil - Exercise books

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Examine likelihood scale - Discuss meaning of each term - Classify statements using likelihood terms - Identify impossible events - Identify certain events - Distinguish between likely and unlikely - Practice with various statements

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Oral questions - Written assignments

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 49 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 11	Lesson: 4
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 5.0: Data Handling and Probability
Sub-strand / lesson focus: 5.2: Probability - Performing chance experiments

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Define chance experiment - Perform chance experiments such as flipping coins, tossing dice, and drawing objects - Show interest in hands-on probability activities

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

What are the possible outcomes when we perform chance experiments?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 210 - Coins - Dice - Colored balls/beads - Bags - Spinners - Recording sheets

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Obtain coins and flip them - Toss dice and record outcomes - Draw colored balls or beads from bags - Use spinners and record results - Record outcomes from experiments - Compare results with other groups - Discuss patterns observed

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Practical tasks - Oral questions

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 50 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 11	Lesson: 5
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 5.0: Data Handling and
Sub-strand / lesson focus: 5.2: Probability - Writing experimental probability outcomes

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain the concept of experimental probability - Write all possible outcomes from chance experiments - Demonstrate systematic recording of outcomes

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we list all possible outcomes from an experiment?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 210 - Coins - Dice - Number cards - Pencil - Exercise books

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - List possible outcomes from coin toss - Write outcomes from die roll - Determine outcomes from spinners - List outcomes from drawing objects - Form combinations of outcomes - Record outcomes systematically - Share findings with class

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Written tests - Problem- solving

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 51 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 12	Lesson: 1
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 5.0: Data Handling and Probability
Sub-strand / lesson focus: 5.2: Probability - Expressing probability outcomes as fractions

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - State the formula for probability as a fraction - Express probability outcomes as fractions accurately - Show understanding of favorable outcomes

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we express the chance of an event happening as a fraction?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 210 - Colored balls/beads - Bags - Calculator - Pencil - Exercise books

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Identify total possible outcomes - Identify favorable outcomes - Express probability as fraction of favorable to total outcomes - Simplify probability fractions - Calculate probabilities from various scenarios - Solve word problems involving probability - Verify answers

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Written assignments - Problem- solving tasks

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 52 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 12	Lesson: 2
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 5.0: Data Handling and Probability
Sub-strand / lesson focus: 5.2: Probability - Expressing probability outcomes as fractions

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - State the formula for probability as a fraction - Express probability outcomes as fractions accurately - Show understanding of favorable outcomes

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we express the chance of an event happening as a fraction?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 210 - Colored balls/beads - Bags - Calculator - Pencil - Exercise books

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Identify total possible outcomes - Identify favorable outcomes - Express probability as fraction of favorable to total outcomes - Simplify probability fractions - Calculate probabilities from various scenarios - Solve word problems involving probability - Verify answers

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Written assignments - Problem- solving tasks

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 53 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 12	Lesson: 3
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 5.0: Data Handling and Probability
Sub-strand / lesson focus: 5.2: Probability - Expressing probability as decimals and percentages

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain the relationship between probability in fractions, decimals and percentages - Convert probability from fractions to decimals and percentages - Demonstrate proficiency in probability conversions

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

Why is probability sometimes expressed as decimals or percentages?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 210 - Calculator - Pencil - Exercise books - Conversion charts

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Convert probability fractions to decimals - Convert probability fractions to percentages - Understand that probability in decimals cannot exceed 1 - Understand that probability in percentages cannot exceed 100% - Calculate complementar y probabilities - Solve problems in different forms - Apply probability in real contexts

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Written tests - Problem- solving

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 54 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 12	Lesson: 4
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 5.0: Data Handling and Probability
Sub-strand / lesson focus: 5.2: Probability - Using IT to play probability games

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Identify digital tools for probability activities - Use technology to play games involving probability and simulate experiments - Appreciate technology in learning probability

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How does technology help us understand probability better?

LEARNING RESOURCES

- MASTER Mathematics Grade 8 Learner's Book pg. 210 - Computers/ta blets - Internet access - Probability apps/software - Projector - Recording sheets

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Access online probability games - Use software to simulate coin flips - Use apps to simulate dice rolls - Play digital probability games - Record results from digital experiments - Compare manual and digital experiments - Discuss advantages of using technology

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Digital portfolio - Practical demonstration - Oral presentation

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 3

LESSON PLAN 55 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 12	Lesson: 5
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: 5.0: Data Handling and Probability
Sub-strand / lesson focus: 5.2: Probability - Using IT to play probability games

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to:

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

LEARNING RESOURCES

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to:

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.